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Your Roll No. .....23BCS066.....

CEN-503

B.Tech.(Computer Engineering) V Semester Examination 2025

Microprocessor

Paper No. : CSC-502/CEN-503(Old)

Time: 3 Hours

Maximum Marks: 45

(Write your Roll No. on top immediately on receipt of this question Paper)

Note: - Attempt all questions by attempting any two parts from each question.

CO1 Q1.

BL2 1(a). Draw and explain Pin Diagram of 8085 microprocessor. 4.5

BL2 1(b). Draw and explain block diagram of 8086 Microprocessor. 4.5

BL2 1(c) Draw and explain timing diagram of LDA 2150H instruction of 8085 stored at address 2000H. 4.5

CO2 **Q2.**

BL2 2(a). Explain the various flags available in 8085 microprocessor with their significance with suitable example.

BL3 2(b).

(i) Explain the function of the following program 1.5

```
        MVI  A , Byte
        ORA   A
        JM    NEXT
        OUT 68H
         HLT
NEXT:  CMA
        ADI 01H
        OUT 68H
        HLT
```

(ii) What is the output at PORT1 when the following instructions are executed? Explain. 1.5

```
MVI  A , 9FH
ADI  82H
JC   Display
OUT  PORT1
HLT
```

```
Display: XRA A
         OUT PORT1
         HLT
```

BL2 2(iii) In the following program explain the range of the bytes that will be displayed at PORT2 1.5

```
         MVI A , BYTE1
         MOV B, A
         SUI 60H
         JC NEXT1
         MOV A , B
         SUI 90H
         JC NEXT2
NEXT1 :  XRA A
         OUT PORT1
         HLT
NEXT2 :  MOV A , B
         OUT PORT2
         HLT
```

BL2 2(c) Explain the meanings of the following instructions of 8085/8086 after writing their syntax. Also illustrate their use with the help of suitable example. 4.5

(i)XTHL

(ii)SPHL

(iii)LOOP

(iv)MOVSB

CO3 Q3.

BL3 3(a). Write a well documented program in the assembly language of 8085 to Perform 8X8 multiplication without shifting multiplicand 4.5

BL3 3(b) Write a program in assembly language of 8085 to arrange 300 integers stored at consecutive memory locations starting with address 2200H in descending order. Program should be well documented. 4.5

BL3 3 (c) Write a program in assembly language of 8085 to generate Modulo-10 counter with a delay of 50 ms between each count assuming a system with 0.5 microsecond clock period. Show all calculations. 4.5

CO4 Q4.

BL2 4(a) Explain the interrupts available in 8085 Microprocessor with the help of suitable diagram. Also explain how multiple interrupts can be implemented using INTR . 4.5

BL2 4(b) Explain how I/O devices are interfaced as I/O Mapped I/O with help of suitable diagram and examples. 4.5

BL2 4(c). What is 8257 DMA controller? Explain with the help of suitable example and diagrams. 4.5

layed 1.5

Q5 Q5.

BL3 5(a). Write a program in assembly language of 8086 to find out the number of numbers and no of negative numbers out of N 16-bit numbers stored at consecutive memory positive locations. 4.5

BL3 5(b). Can assembly language be used with C programming? If yes discuss how it can be done with suitable example. Also discuss how are software interrupt used in programming with suitable example. 4.5

BL3 5(c). Write a program in assembly language of 8086 to convert a 16-bit binary number into equivalent BCD number. 4.5